

₹50,000 devices with acceptable error margins, which still ensure no critical condition goes undetected.

How do you support hardware startups? Any early challenges in prototyping or validation?

We offer electronic design automation (EDA) tools, connect startups to early customers, and provide board design and prototyping support through IIT Madras and global partners such as Taiwan Semiconductor Manufacturing Company Limited (TSMC) via Multi-Project Wafer (MPW) slots. Given the high costs, we use vendor tie-ups, MoUs with institutions, and shared infrastructure to ease access. We offer startups access to advanced labs, prototyping tools, and early validation via industry tie-ups—especially vital for sensor or embedded system ventures. Earlier, hardware development was limited by ecosystem gaps, such as lack of access to high-speed ASICs and reliance on outdated components. Even with strong design skills, projects like the Shakti processor faced performance limits due to subpar IPs.

How does your shared IP and tool licensing model work? Any regulatory hurdles?

We centrally license tools and share them with startups, cutting costs while ensuring ethical use and vendor trust. As a Section 8 non-profit, Pravartak focuses on social impact, not profit. It supports open source and commercial IP through legal safeguards. Hurdles such as foreign payments and procurement policies require government support.

How do you collaborate with startups and industry to ensure your solutions are deployment-ready?

We follow a dual-track approach. First, we leverage the strength of startups incubated at IIT Madras, the Research Park, and Pravartak. When a customer approaches us with a specific requirement, such as the government's Department of Defence needing portable medical machines,

we evaluate our startup ecosystem and connect the right players. In one case, we linked two startups to co-develop a solution. The second approach is when the requirement demands new research. In such cases, we initiate collaborative projects through IIT Madras, particularly via the Centre for Industrial Consultancy and Sponsored Research (IC&SR). We know which professors specialise in which areas, and we bring them into the loop. Once a solution is developed, our team validates it and delivers it to the customer.

How do you balance government grants with private funding, and what is the road ahead?

There is often a disconnect; industry funds R&D for real outcomes, but academia often stops at papers. That is why under the National Mission on Interdisciplinary Cyber-Physical Systems (NM-CPS) mission, we focus on translational research—taking technology from lab to market. Our model blends Department of Science and Technology (DST) grants with matching private/public funding, ensuring relevance and accountability. With ₹3 billion already raised and financial independence achieved, we are now eyeing international expansion via IITM Global, taking startups and technology abroad, not just to export but to collaborate.

Could you share examples where research incubated turned into commercial success?

A strong example of our deep-tech incubation success is TuTr Hyperloop, born out of the student-led Avishkar Hyperloop project, which recently secured its first customer, Karaikal Port. MindGrove, focused on affordable electronics manufacturing, is actively engaging with potential customers. While deep-tech startups usually take five to six years to mature and Pravartak itself is just over four years old, we are starting to see solid traction. The model works when startups,

researchers, and industry align with intent and persistence.

You mentioned DST allows collaboration beyond IIT Madras. Could you share more about those partnerships?

Yes, we are not restricted to IIT Madras. DST has given us the flexibility to work with any Indian institution. We collaborate with institutes such as IIITDM Kancheepuram, IIT Kanpur, IIT Mandi, IIT Hyderabad, and private institutions such as MIT Art, Design & Technology University (MIT ADT) in Pune. We even work with regional universities and remote colleges—for example, an arts college in Karur. We aim to discover and nurture talent across the country, not just in Tier-1 institutions.

How do IIT Madras and DST support your innovation and partnerships?

IIT Madras is a foundational pillar for us. Around 60-70% of our solutions originate from its research ecosystem. We tap into startups incubated there, leverage faculty expertise, and access cutting-edge labs. It gives us a strong, credible pipeline for deep-tech problem-solving.

What is your 3-5 year roadmap, and long-term vision for India in electronics and deep-tech?

We are focusing on two major areas: aviation and electronics, with a special emphasis on safety and cybersecurity. In aviation, the goal is to promote safe aviation engineering. In electronics, we are zeroing in on secure components and embedded cybersecurity. Given the rise in cyber threats and the strategic importance of electronics, this is a natural direction for us.

Looking five years ahead, what is your vision for IITM Pravartak?

We aim to see at least four to five unicorns emerge from Pravartak-supported startups in the next five years. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/interviewNov2025>

